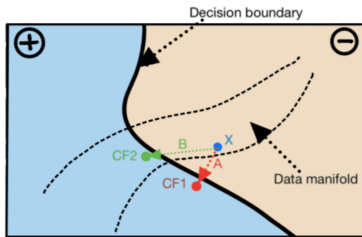
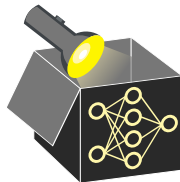


Interpretable Machine Learning

Counterfactual Explanations (CEs) Optimization Problem and Objectives



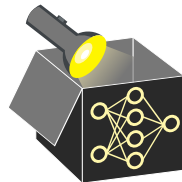
Learning goals

- Formulate CEs as optimization problem
- Identify key objectives (proximity, sparsity)
- Understand trade-offs in CE generation

MATHEMATICAL PERSPECTIVE

Terminology:

- $\mathbf{x}$: original/factual data point whose prediction we want to explain
- $y' \subset \mathbb{R}^g$: desired prediction ($y' = \text{"grant credit"}$) or interval ($y' = [1000, \infty[$)



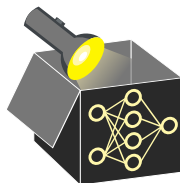
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A **valid** counterfactual $\mathbf{x}'$ satisfies two criteria:

- 1 **Prediction validity:** CE's prediction $\hat{f}(\mathbf{x}')$ is equal to the desired prediction y'
- 2 **Proximity:** CE $\mathbf{x}'$ is as close as possible to the original input $\mathbf{x}$



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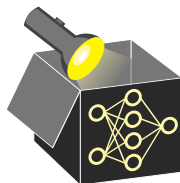
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Reformulate these two objectives as optimization problem:

$$\arg \min_{\mathbf{x}'} \lambda_1 o_{target}(\hat{f}(\mathbf{x}'), y') + \lambda_2 o_{proximity}(\mathbf{x}', \mathbf{x})$$

- λ_1 and λ_2 balance the two objectives
- o_{target} : distance in target space
- $o_{proximity}$: distance in feature space



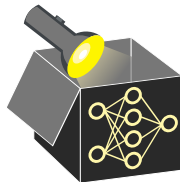
Distance in target space O_{target} :

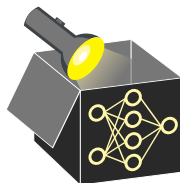
- **Regression:** L_1 distance

$$O_{target}(\hat{f}(\mathbf{x}'), y') = |\hat{f}(\mathbf{x}') - y'|$$

- **Classification:**

- For predicted probabilities: $O_{target}(\hat{f}(\mathbf{x}'), y') = |\hat{f}(\mathbf{x}') - y'|$
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Distance in input space $O_{proximity}$: Gower distance (mixed feature types)

$$O_{proximity}(\mathbf{x}', \mathbf{x}) = d_G(\mathbf{x}', \mathbf{x}) = \frac{1}{p} \sum_{j=1}^p \delta_G(x'_j, x_j) \in [0, 1]$$

where

- $\delta_G(x'_j, x_j) = \mathbb{1}_{\{x'_j \neq x_j\}}$ if x_j is categorical

- $\delta_G(x'_j, x_j) = \frac{1}{\widehat{R}_j} |x'_j - x_j|$ if x_j is numerical

$\rightsquigarrow \widehat{R}_j$: range of feature j in the training set to ensure $\delta_G(x'_j, x_j) \in [0, 1]$

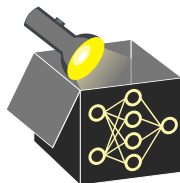
FURTHER OBJECTIVES: SPARSITY

Additional constraints can improve the explanation quality of the corresponding CEs

↔ popular constraints include **sparsity** and **plausibility**

Sparsity Favor explanations that change few features

- End-users often prefer short over long explanations



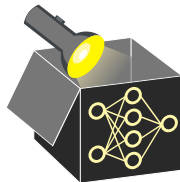
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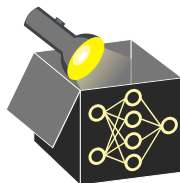
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e.g., using L_0 -norm (number of changed features) or L_1 -norm (LASSO)
- Alternative: Include separate objective measuring sparsity, e.g., via L_0 -norm

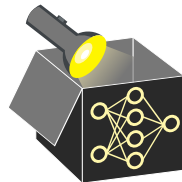
$$o_{sparse}(\mathbf{x}', \mathbf{x}) = \sum_{j=1}^p \mathbb{1}_{\{x'_j \neq x_j\}}$$



FURTHER OBJECTIVES: PLAUSIBILITY

Plausibility:

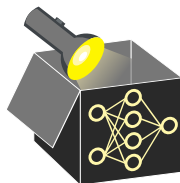
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 $\rightsquigarrow$ Implausible: increase income *and* become unemployed



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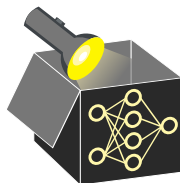
- CEs should suggest realistic (i.e., plausible) alternatives
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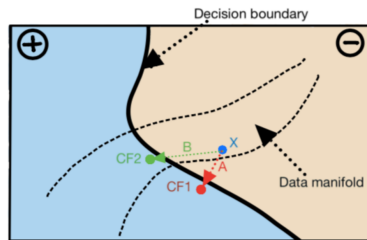
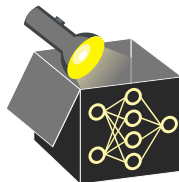
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Example from ▶ "Verma et al." 2020

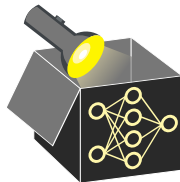
- Input $\mathbf{x}$ originally classified as $\ominus$
- Two valid CEs in class $\oplus$: CF1 and CF2
- Path A (CF1) is shorter (but unrealistic)
- Path B (CF2) is longer but in data manifold

FURTHER OBJECTIVES

Plausibility term: Encourage counterfactuals close to observed data.

- Define $\mathbf{x}^{[1]}$ as the nearest neighbor of $\mathbf{x}'$ in the training set $\mathbf{X}$
- Use Gower distance between $\mathbf{x}'$ and $\mathbf{x}^{[1]}$ to define plausibility objective:

$$o_{plausible}(\mathbf{x}', \mathbf{X}) = d_G(\mathbf{x}', \mathbf{x}^{[1]}) = \frac{1}{p} \sum_{j=1}^p \delta_G(x'_j, x_j^{[1]})$$



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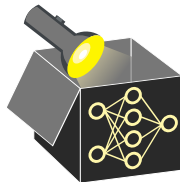
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Extended optimization: Add sparsity and plausibility terms to the objective

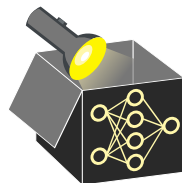
$$\arg \min_{\mathbf{x}'} \lambda_1 o_{target}(\hat{f}(\mathbf{x}'), y') + \lambda_2 o_{proximity}(\mathbf{x}', \mathbf{x}) + \lambda_3 o_{sparse}(\mathbf{x}', \mathbf{x}) + \lambda_4 o_{plausible}(\mathbf{x}', \mathbf{X})$$



REMARKS: THE RASHOMON EFFECT

Issue (Rashomon effect):

- Solution to the optimization problem might not be unique
- Many equally close CEs might exist that obtain the desired prediction
⇒ Many different equally good explanations for the same decision exist



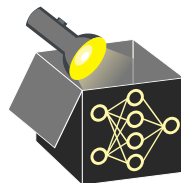
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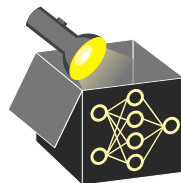
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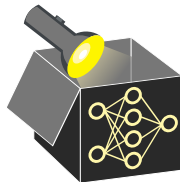
Note:

- Nonlinear models can produce diverse and inconsistent CEs
↔ suggest both increasing and decreasing credit duration
(confusing for users)
- Handling this **Rashomon effect** remains an open problem in interpretable ML



REMARKS: MODEL OR REAL-WORLD

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~> Transfer of model explanations to explain real-world is generally not allowed
- **Example:** CE suggests increasing age by 5 to receive a loan
~> The applicant waits 5 years and reapplies



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- **Example:** CE suggests increasing age by 5 to receive a loan
~> The applicant waits 5 years and reapplies
- **Problem:** Other features may change till then (e.g., job status, income)
~> ▶ "Karimi et al." 2020 propose CEs that respect causal structure
- **Model drift:** Bank's algorithm itself may change over time
~> Past CEs may become invalid

